MAYUKH BANERJEE

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SUMMARY

AI Engineer with production experience across agentic systems, RAG pipelines, and LLM orchestration (LangChain, LangGraph) — shipped across a live B2B SaaS ecosystem and published research. Seeking AI/ML Engineering roles owning systems end-to-end: architecture, prompt design, production.

TECHNICAL SKILLS

Languages	Python, JavaScript, TypeScript, C++, Java
AI / ML	LangChain, LangGraph, MCP (Model Context Protocol), Agentic AI, RAG, Schema-Conditioned Generation, Context Engineering, Prompt Engineering, OpenAI API, Groq API, Gemini API, Hugging Face, PyTorch, Keras, scikit-learn, XGBoost
Backend	FastAPI, Node.js, Express.js, Pydantic, REST APIs, Next.js
Vector / Data	Qdrant, FAISS, Chroma, Pinecone, DynamoDB, NumPy, Pandas, Oracle SQL
Cloud & Tools	AWS (Lambda, S3, Cognito, Transcribe, Comprehend, Amplify, CloudWatch), EnergyPlus, Streamlit, HeyGen, ElevenLabs, Git, GitHub, Google Apps Script

EXPERIENCE

Bengaluru, India **May 2026 – Present**

AI Engineer (Intern), Easehawk Technologies

Owned AI across 5 live products: Moxsend (B2B outreach), MoxReply (inbox automation), MoxCRM (lead management), MoxCampus (EdTech), EaseHawk Avatar Chatbot.

- **Cut LLM API costs 90%+** on a live client-facing outreach pipeline — replaced per-lead calls with cohort-based orchestration (one master template, dynamic merge tags per segment).
- **Personalized outreach for real B2B clients** with a dual-context RAG pipeline (LangChain + Qdrant) blending lead profile and campaign brief — no two emails are the same template.
- **0% URL hallucinations** in the knowledge layer, achieved by converting raw PDFs and live sites into dense fact summaries before they touch the prompt; shipped with Arabic localization for GCC markets.
- **Eliminated pre-LLM classification cost entirely** with a zero-token deterministic engine sorting leads across 6 sectors and 20+ roles at 100% accuracy.
- **Shipped MoxCampus AI to real schools** (FastAPI + LangChain) — 4 report types per student, plus a HeyGen-powered avatar guide with synced voice navigation.

Vellore, India **2025**

Technical & Research Team Lead *E-Cell, VIT*

- **Zero downtime across 8 hours** automating 100+ live data sheets (Python + Google Apps Script) for a flagship event — 5,000+ applicants, 200+ participants, Rs. 1L prize pool.

PROJECTS

Evoke — Talk to the People You Have Lost – *Schema-Conditioned RAG, AWS Serverless, AWS Bedrock, LangChain* VIT, 2026 | Research Paper

Grief doesn't come with an off switch. Evoke keeps a loved one's voice, humor, and instincts one call away.

- **Cut confident fabrication from 15.3% to 2.0%** (Cohen's d = 1.31, $p < .001$) with a schema-conditioned RAG pipeline and an epistemic-humility gate that blocks the model from inventing opinions the person never held — authenticity scores up 1.47 points over a standard LLM baseline.
- **Consent enforced at the infrastructure level** across a 10-service AWS serverless pipeline (Lambda, S3, DynamoDB TTL, Cognito, Transcribe, Comprehend, Amplify, CloudWatch) — data expiry and access control are architectural invariants, not policy. Runs at $0 marginal cost within AWS Free Tier.

EcoLoop — AI That Controls Buildings, Not Just Software – *MCP, LLM Orchestration, EnergyPlus, Groq, Streamlit, Python* VIT, 2026

Buildings burn 30–40% of their energy on HVAC and lighting running blind on static schedules, not who's actually there.

- **33.8% electricity reduction** (114.1 vs. 172.5 kWh) and a **10.1-point comfort gain** against a physics-identical uncontrolled baseline — every number pulled directly from simulation artifacts.
- **Full MCP tool-calling architecture** (5 Pydantic-validated tools): writes queue until end-of-timestep so a deterministic clamp layer enforces safety before any value reaches the actuator. The prompt is advisory. The clamp is law.

Scheduler AI — Find the Right Service Provider, Instantly – *LangGraph, Agentic AI, FastAPI, Next.js, MongoDB* VIT, 2026

Small providers struggle to be found; customers struggle to find someone reliable at a fair price — Scheduler AI fixes both sides.

- **Zero hallucinated bookings by design**: a 6-node LangGraph StateGraph confines the LLM to 2 boundary roles while all search, ranking, and slot verification run as deterministic code.

LeBy — The Law in Plain English – *RAG, FAISS, Gemini API, LangChain, FastAPI, React* VIT, 2026

Understanding your rights shouldn't require a law degree — LeBy turns the Constitution and case law into plain English.

- **Plain-language legal answers in under 5 seconds**: a FAISS-backed RAG pipeline (Gemini API) classifies Indian legal documents across 4 domains with near-zero hallucinations, full-stack FastAPI + React.

EDUCATION

Vellore, India **Aug 2023 – Jun 2027**

Vellore Institute of Technology *B.Tech, Information Technology | GPA: 8.56/10*

CERTIFICATIONS

Machine Learning A-Z: AI, Python & R | Complete Python Developer | IBM DevOps & Agile Designs | AWS Cloud Architect